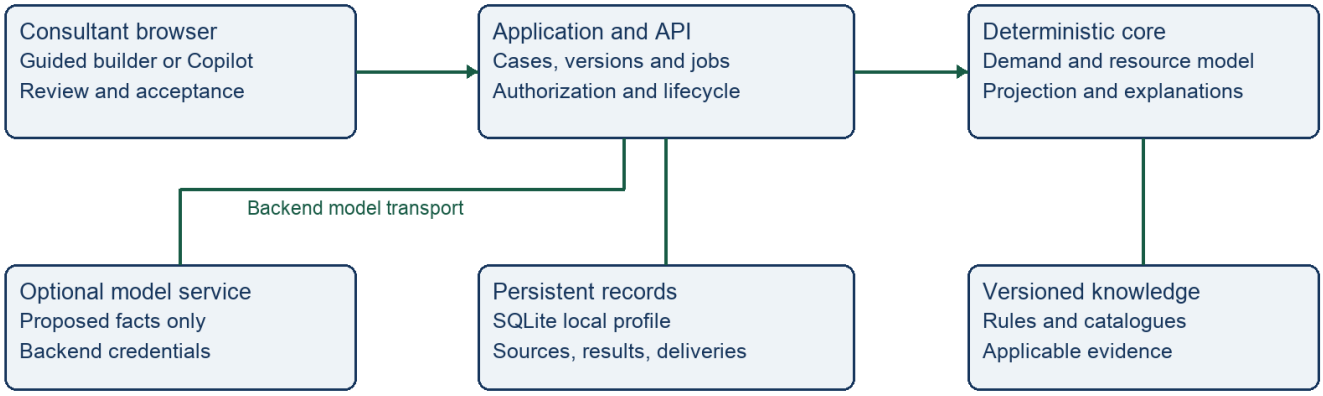
Anubis technical solution architecture

SME evaluation edition 1.0 | 2 November 2026

System purpose and boundary

The application provides a local, API-driven consultation workspace around a deterministic workload compiler. Its output is a workload requirement and design handoff, not an approved bill of materials. The optional model service assists discovery. It cannot accept its own proposals or replace the sizing engine.



End to end transformation

The consultant starts with either an architecture builder or a Copilot description. Both routes converge on structured workload data. Reviewed facts retain their origin and units. A frozen revision and selected knowledge define the calculation. Results are immutable. A projection maps a selected candidate into the existing downstream contract and records omitted requirements in a companion specification.

Completion snapshots retain the selected result and projection. If supported sizing is impossible, report-only closure consolidates the evidence and unresolved work. Reopening starts a new editing cycle while retaining previous deliveries. Recoverable deletion moves an engagement to Trash rather than erasing its history.

Authority and trust boundaries

Boundary	Permitted action	Control
Customer description to fact proposal	Extract candidate architecture and quantities	Schema validation, exact source references and explicit unknowns
Proposal to accepted workload	Human accepts a coherent proposal	Current draft version and acceptance of visible assumptions
Workload to result	Deterministic compilation	Frozen input, versioned catalogues and numerical validation
Result to design projection	Map a selected candidate	Supported fields, explicit substitutions and mapping ledger
Projection to delivery	Human records delivery	Confirmation, current versions and immutable snapshots
Application to external service	Provider call or approved tool access	Backend credentials, origin policy, budgets and explicit retry

User experience architecture

The four engagement stages are description, workload characterization, calculation and design handoff. The guided editor has its own four substeps: Choose application, Confirm arrangement, Describe scale, Review and estimate. These are nested intake steps rather than four additional engagement phases.

Architecture diagrams show customer traffic, local compute, outside services and data boundaries. A diagram helps locate measurements. It does not prove that all depicted relationships have a quantitative simulator behind them. Each pattern exposes its quantitative coverage and required evidence.

The Copilot page distinguishes the description box, active request status, proposed facts and the next action. Acceptance leads to Review and calculate GPU estimate. Calculation leads to Continue to Design handoff. Delivery leads to Saved deliverables. A completed Copilot session is historical and read-only until the engagement is reopened.

Cases show current phase, next action and controls in a bordered table. Case lifecycle and engineering readiness are separate. An open case can contain a successful calculation. A completed case records delivery. A report-only closed case records a conclusion without a valid handoff.

Data architecture

Record	Purpose	Relationship and persistence
Case and membership	Engagement name, access and lifecycle	Parent scope for drafts, jobs, sources and deliveries
Draft and frozen revision	Editable workload then immutable calculation input	Version checks prevent silent stale overwrite
Copilot session and sources	Description, uploaded evidence and turn history	Scoped to case and draft, with content digests
Fact proposal and review	Structured facts, original units, status and source quotation	Acceptance translates facts through deterministic rules
Knowledge pack and activation	Rules, model assumptions and applicable evidence	Version and digest accompany interpretation or compilation
Job and result	Durable execution state and immutable numerical output	Job success does not imply an engineering pass
Projection and assessment	Existing target intake and companion requirements	Linked to the exact selected result and target contract
Completion or closure report	Delivered snapshot or unresolved final report	Historical record survives reopening

SQLite is the local persistence profile. The application stores structured records and artifacts in the database. Private identities are separate from the database. Backup must preserve the database consistently and protect identity material separately. Experimental PostgreSQL support is not a qualified shared-service deployment.

Deterministic engine and knowledge

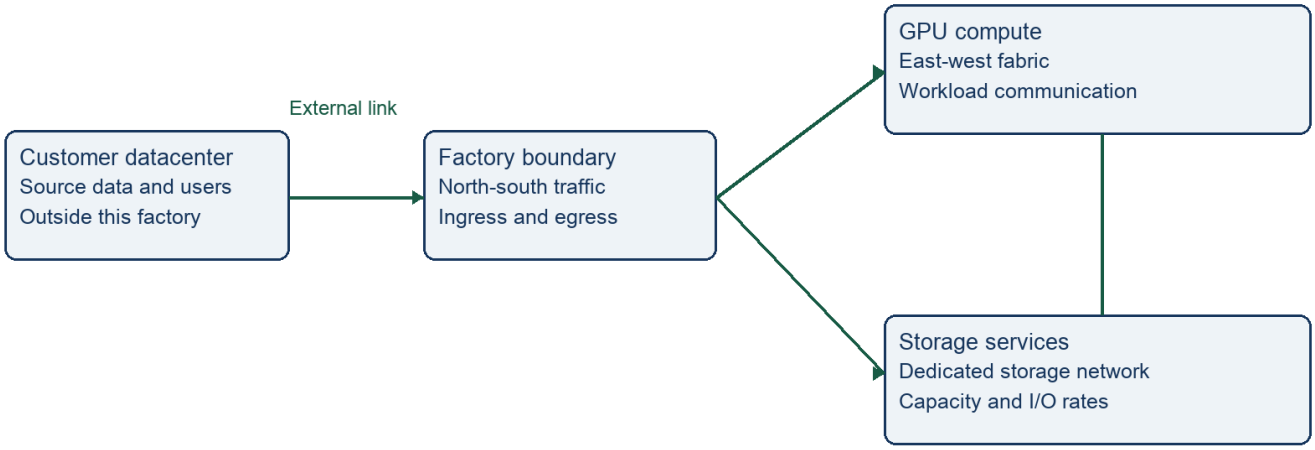
The compiler validates dimensions, derives demand, resolves memory and throughput constraints, evaluates operating scenarios and produces explanations. Analytical planning applies explicit effective-compute and memory-bandwidth assumptions. Exact rational quantities are used internally, with formatted result strings at presentation boundaries. Some downstream calculations consume displayed values, so precision and boundary cases still require regression checks.

The CKP and EKB concept is useful as a governance distinction. CKP represents executable contracts, rules and catalogued assumptions. EKB represents attributed observations and applicability constraints. The current repository implements this through versioned JSON catalogues, evidence records and

activation controls. It does not implement a general knowledge-graph platform or a universal autonomous research pipeline.

Knowledge family	Current implementation	Maintenance requirement
Architecture and discovery	architecture-core.json, guided-intake.json, workload-interpretation.json	Pattern definitions, questions, scope and diagrams must remain consistent
Planning models and hardware	planning.json and compiled planning_catalog.inc	Synchronize generated catalogue, units, defaults and version
Performance evidence	Public and fixture evidence packs	Retain configuration, source, date, applicability and synthetic labels
Design mapping	Projection adapter and target contract	Preserve target constraints, omissions and substitution decisions
Validation knowledge	Scenario fixtures and independent oracles	Separate expected behaviour from empirical performance claims

Four network roles



The customer datacenter connection carries initial and ongoing external transfers. North-south is the factory border path for that traffic. Do not add those two descriptions as independent copies of the same bytes. The storage network is physically dedicated. East-west is the GPU communication fabric. Missing throughput is not inferred from storage capacity or from a topology name.

Copilot integration

A provider profile identifies origin, API path, model, backend secret variable, output-token budget, turn budget, data policy and optional reasoning setting. Public adapters cover OpenAI, Anthropic, DeepSeek and xAI. A compatible corporate gateway is supported. A corporate MCP model tool needs the explicit model-tool contract. A discovery-only MCP server is not a substitute for an inference service.

Fact extraction is probabilistic. Normalization, unit conversion, architecture matching, validation, review, compilation and projection are deterministic. A response may receive one bounded correction attempt. Partial recovery is disclosed and does not count as a fully successful interpretation. There is no silent provider fallback.

Read-only MCP tools require approved names and schema digests. The external stdio MCP facade exposes scoped operations with case budgets. It does not give agents authority to approve facts, manage credentials, publish evidence or submit a design. Advanced manual changes are preserved when disjoint, and conflicts are blocked before a write.

Deployment and security

The supported evaluated route binds to loopback. Access tokens bootstrap private local identities and browser sessions. Case membership is enforced even for administrators. If-Match protects concurrent editing. Idempotency controls protect supported retried mutations. Model credentials are resolved in the backend and must not enter customer exports.

Nonstandard origins require an administrator's exact origin allowlist. Transport rejects redirects and applies address checks. Public-only profiles cannot process corporate-classified sessions. Cancellation suppresses publication but cannot guarantee that a provider stops billed work. Pending external outcomes after a restart require inspection and explicit retry or reconciliation.

The local profile is not ready for exposure as a multi-organization Internet service. Federation, session expiry, rate limiting, multi-host isolation, automated transcript expiry, tested disaster recovery objectives and a qualified schema migration process remain outside the delivered production scope.

Interface and handoff contracts

The authenticated OpenAPI description and operations catalogue are the authoritative API inventory. Existing downstream adapter identifiers remain for compatibility even though customer-facing text is tool-neutral. Generic exports can support another architect or tool. They do not imply automatic API compatibility with arbitrary design products.

The handoff contains the supported intake payload, companion requirements, mapping ledger and deterministic executive summary. The installed adapter can perform an explicitly approved experimental local submission and preserve the returned assessment. A lost external response must not trigger an unreviewed duplicate submission.

Evaluation decisions

Accept the local solution architecture if both user routes converge on the same workload semantics, the provider cannot bypass review, every result can be traced to versions and closure preserves historical deliverables. Treat model quality and physical performance as separate qualification activities. See documents 06 through 08 for implementation mapping and executable checks.

Release implementation addendum — 3 November 2026

This addendum supersedes earlier descriptions of unsupported specialist sizing and result interpretation. The original paper remains the conceptual basis. Author: Traiano Giuseppe Welcome. Contact: traiano@gmail.com.

GPU-complete planning scenarios

All sixteen demonstration scenarios now contain accepted inputs sufficient for positive B300, B200 and H200 workload GPU estimates. These are planning examples. They do not establish measured hardware performance. Specialist examples declare platform-specific reference throughput, per-GPU memory, complete GPU-group size and independent group replication assumptions.

Reference capacity and uncertainty

For independent reference groups, group count is the ceiling of aggregate demand divided by capacity per complete group. The calculation preserves complete groups, applies deployment granularity and checks per-GPU memory. Platform throughput is supplied explicitly, not inferred from another GPU. Sensitivity envelopes are planning variations, not statistical confidence intervals. Workload GPU counts remain distinct from factory deployment anchors.

Explain this to me

A read-only explanation is available from saved results and Design handoff, including incomplete or unsuccessful outcomes. The deterministic engine supplies the facts. An optional configured model adds brief interpretation around immutable fact placeholders. Six sections cover outcome, compute, memory, storage, networks and next steps. Missing requirements remain missing. Outside services do not silently become local retained data or initial transfer requirements.

Controls and persistence

The explanation packet references an exact result and, where supplied, a matching projection. Provider access follows existing classification and connection policy. The model cannot invoke sizing or editing operations. The server validates section structure and fact placeholders, rejects extra numerical claims, allows one bounded correction and falls back to deterministic prose on failure. Qualitative model prose remains interpretation and requires consultant review. Explanations are stored separately with provenance and can be retrieved and downloaded without changing results.

Verification and evidence

The repository contains repeatable fixture tests for all sixteen scenarios, incomplete outcomes, invented quantities, access control and projection matching. Live DeepSeek explanation validation produced model-assisted explanations for all sixteen saved results, with unchanged calculation records. See testing-evidence/explanations. Four illustrated Word playbooks cover both manual and assisted workflows. Hardware qualification and universal correctness of model prose are not established by these checks.